

Tests of numerical errors between Matlab and Python functions

This document is a report of all the numerical errors that were measured between Numpy/Scipy and Matlab's functions for sound processing .

First : Interpolations :

The Outer Middle Model contains two steps of interpolations , the two interpolated lists are the added in tfOuterMiddle, here is the error resulting of the same computation between Matlab and Numpy's interpolation.

1 : Outer Middle Model :

Here are the measures done on the list called OuterMiddle of the data_glasberg matlab file reproduced in python.

OuterMiddle Filter

! Possibly irrelevant measures due to mismatch of size between the lists,

Length of python list : 15980

Length of matlab list : 15981

Maximum error for outerMiddle is : 1.7343616127996384

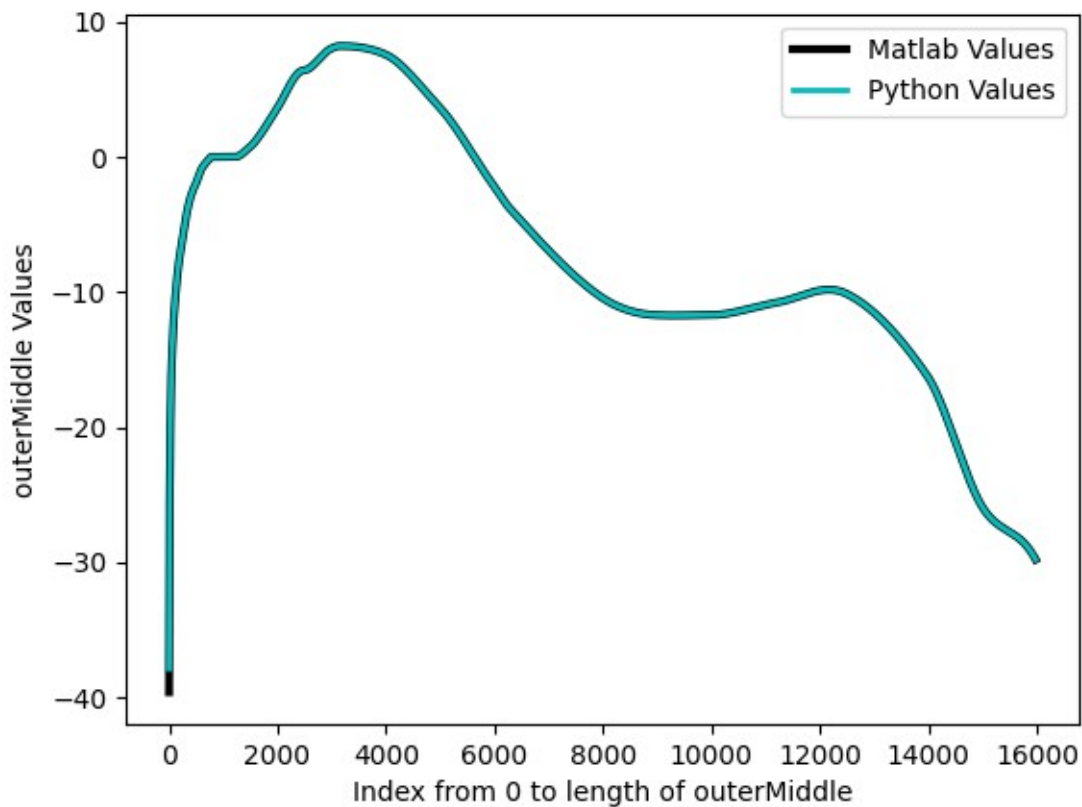
Average error for outerMiddle is 0.005610347542319035

Quadratic error for outerMiddle is : 0.0014016576297613225

Percentage of error : ratio = (max_error / average_MatlabOuterMiddle) is : 3.628141092298088 %

Quadratic Percentage of error : ratio = (Quad_error / range of values) is : 0.0029321518686412974 %

max for matlab : 8.20303656 , max for py : 8.203036556006735
32001



Values of a

! Possibly irrelevant measures due to mismatch of size between the lists,

Length of python list : 149

Length of matlab list : 55

Maximum error for a is : 4.8970694166428075e-09

Average error for a is 1.6499873144549715e-09

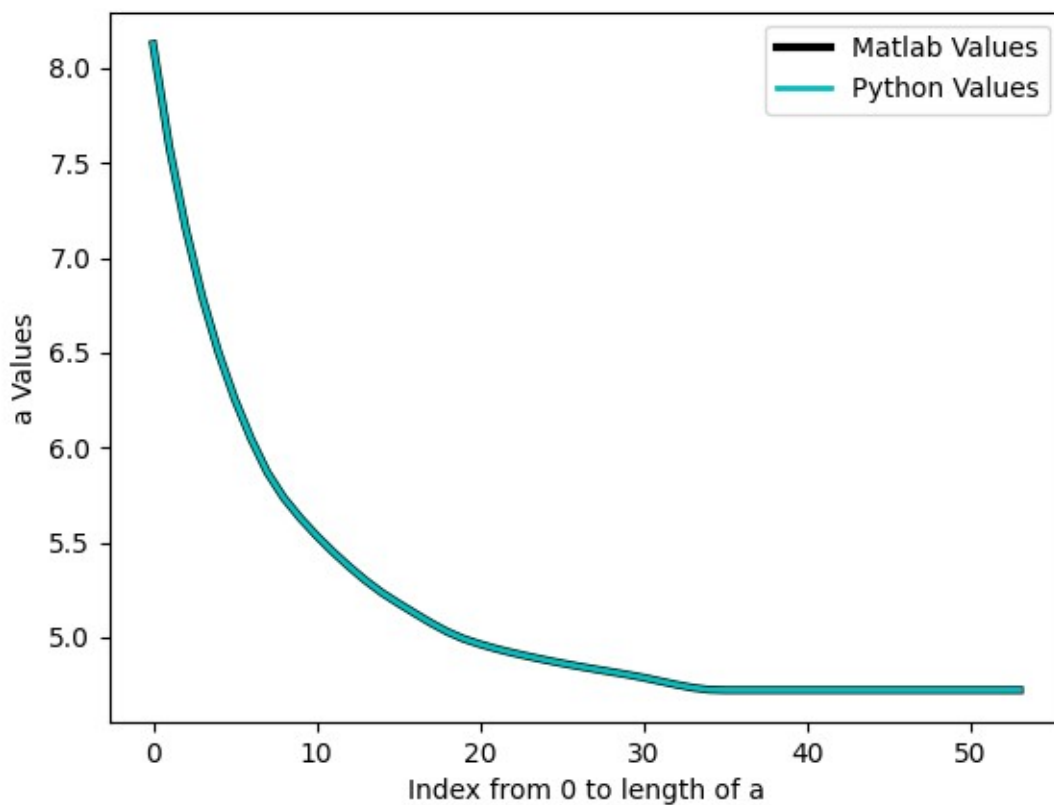
Quadratic error for a is : 5.717956564795287e-18

Percentage of error : ratio = (max_error / average_MatlabOuterMiddle) is :
1.437688075908633e-07 %

Quadratic Percentage of error : ratio = (Quad_error / range of values) is :
1.6786852038142726e-16 %

max for matlab : 8.12717133 , max for py : 8.127171332418412

55



tQ values

! Possibly irrelevant measures due to mismatch of size between the lists,

Length of python list : 149

Length of matlab list : 55

Maximum error for tQ is : 4.909670678898692e-07

Average error for tQ is 2.306364883317904e-08

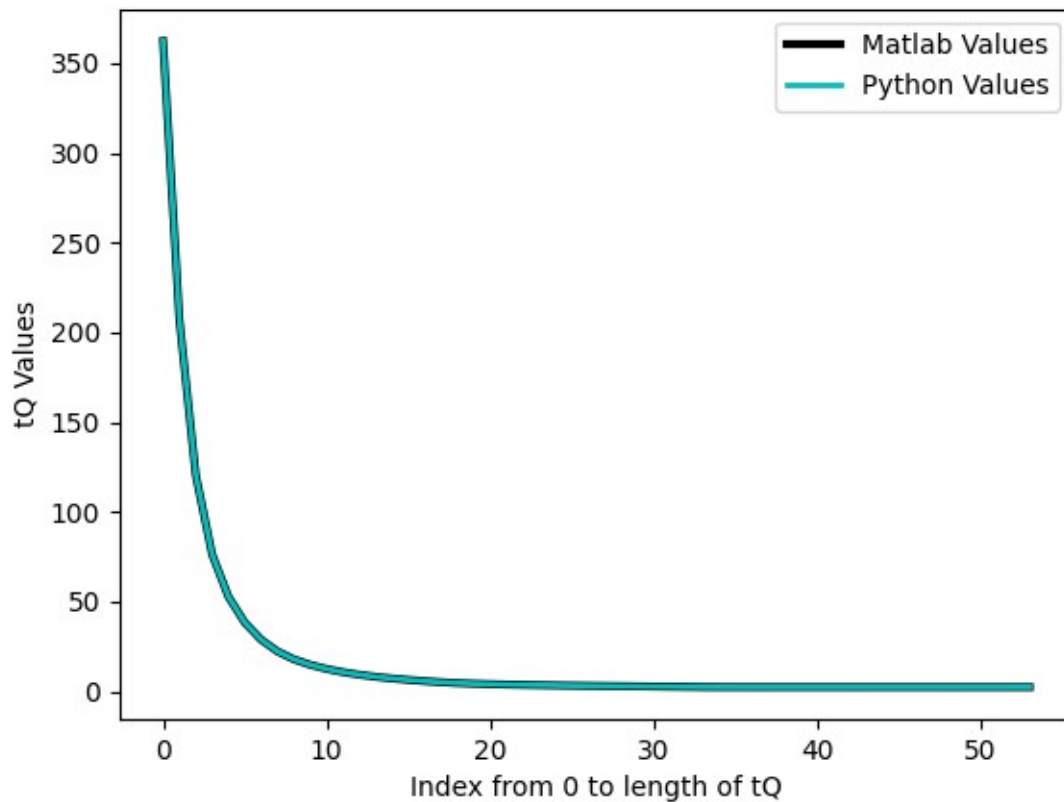
Quadratic error for tQ is : 6.714659876835601e-15

Percentage of error : ratio = (max_error / average_MatlabOuterMiddle) is :

1.3637870241653333e-07 %

Quadratic Percentage of error : ratio = (Quad_error / range of values) is : 1.865169093941371e-15 %

max for matlab : 362.36322 , max for py : 362.3632198891883



FiltFilt

! Possibly irrelevant measures due to mismatch of size between the lists,

Length of python list : 32000

Length of matlab list : 32001

Maximum error for filtfilt is : 0.00027648026561593247

Average error for filtfilt is 8.825674773514864e-05

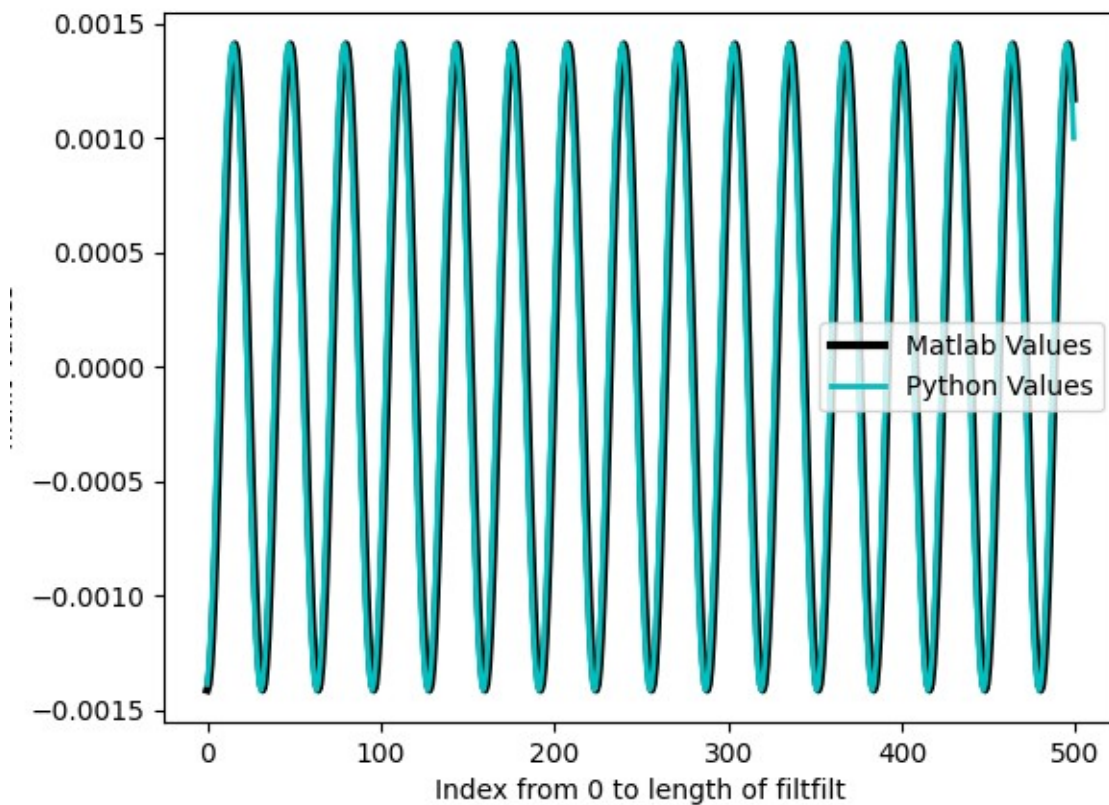
Quadratic error for filtfilt is : 1.2792098681213893e-08

Percentage of error : ratio = (max_error / average_MatlabOuterMiddle) is : 9.151191089394118 %

Quadratic Percentage of error : ratio = (Quad_error / range of values) is :

0.00042340432220500936 %

max for matlab : 0.00151062448 , max for py : 0.0014121681908427974



With the initial values of the model done, we can now test values that are relative to a signal.

Signal

Maximum error for signal is : $4.999384803708318 \times 10^{-12}$
Average error for signal is $1.3585596463020968 \times 10^{-12}$
Quadratic error for signal is : $4.1735019715012936 \times 10^{-24}$

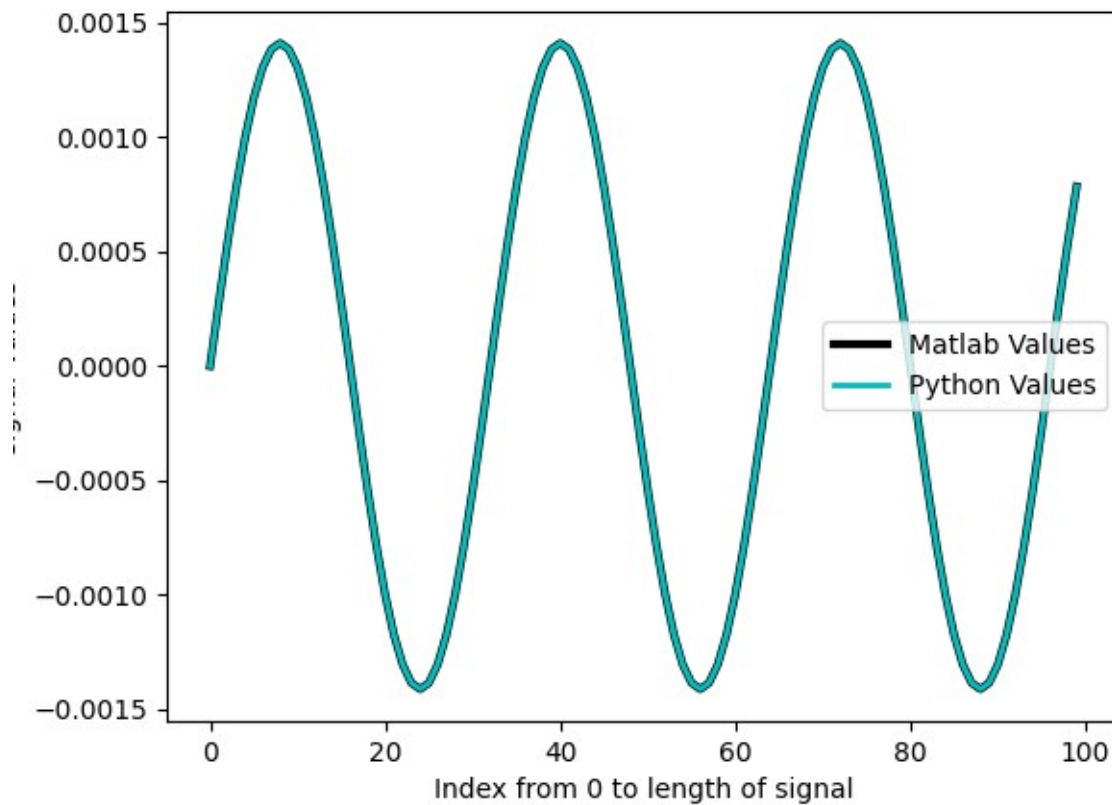
Percentage of error : ratio = (max_error / average_MatlabOuterMiddle) is :

1.769646700281891e-07 %

Quadratic Percentage of error : ratio = (Quad_error / range of values) is : 1.477306565201564e-19 %

max for matlab : 0.00141253754 , max for py : 0.0014125375429208428

15981



Elapsed time for constructor: 0.021743059158325195

fftlen : 32000

Elapsed time : 6.715368747711182

149

Excitation patterns

Maximum error for eLdB is : 0.003487367101419281

Average error for eLdB is 0.002259505113780775

Quadratic error for eLdB is : 5.23546413429215e-06

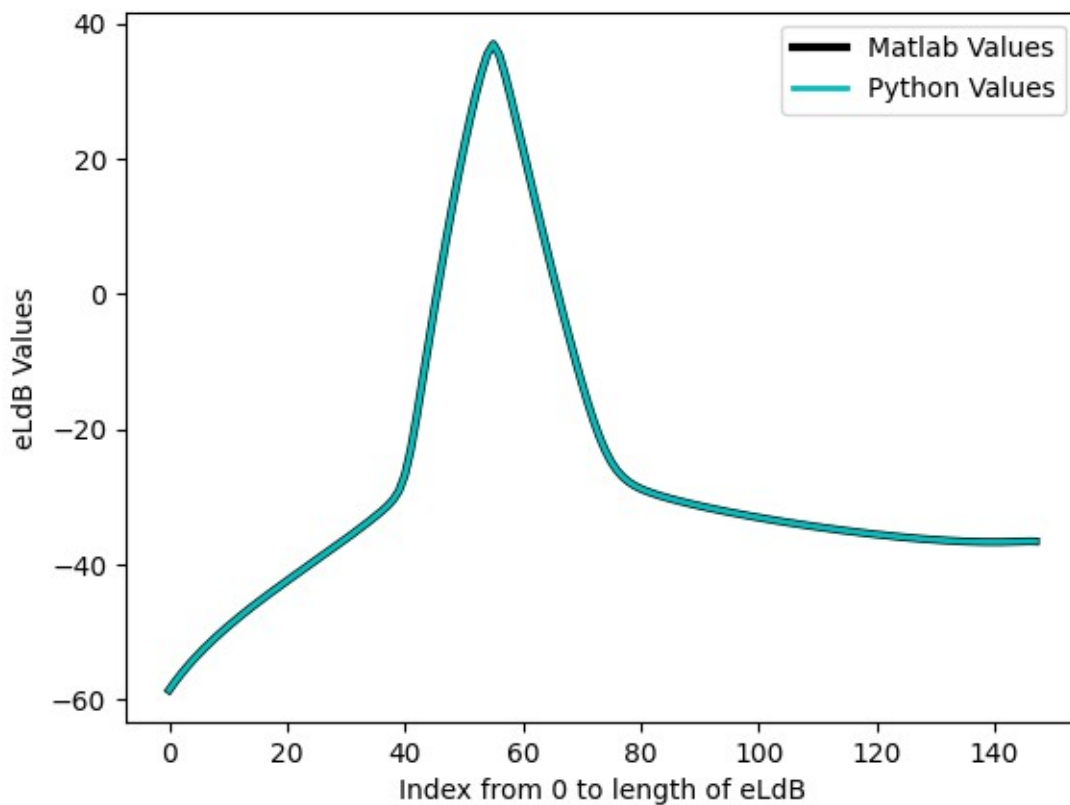
Percentage of error : ratio = (max_error / average_MatlabOuterMiddle) is :

0.0036504651503218733 %

Quadratic Percentage of error : ratio = (Quad_error / range of values) is : 5.480317618473679e-06 %

max for matlab : 36.9729305 , max for py : 36.97520218248139

149



SpecLoudness

Maximum error for specLoudness is : $4.846299277856553e-18$

Average error for specLoudness is $3.9363827747873567e-20$

Quadratic error for specLoudness is : $1.6270234748723158e-37$

Percentage of error : ratio = (max_error / average_MatlabOuterMiddle) is : 40.89831137044033 %

Quadratic Percentage of error : ratio = (Quad_error / range of values) is :

$1.3730582629592483e-18$ %

max for matlab : $1.18341462e-17$, max for py : $1.1849631721859461e-17$

55

